LLM: SFT

SFT (Supervised Fine-Tuning): modelni savollarga qanday tugri javob berish va kursatmalarga amal qilish uchun savol-javob misollari orqali base modelni qushimcha uqitish.

- Supervised: har bir sample uchun tugri javob (label) beradi
- Fine-Tuning: mavjud vaznlarni kichikroq LR bilan ozgina uzgartiradi

Loss: pre-training bilan SFT uchun lossni ozgina farqi buladi.

Loss	**Pre-Training**	**SFT**
Loss type	Cross-entropy	Cross-entropy
Data	Raw text	Instruction
Qaysi tokenlarda ishlaydi	Barcha tokenlarda	Faqat response tokenlarda
Boshlanish	Random weights	Pre-trained model weights
Size	Billions of tokens	1k-50k juftlik

SFT nimani urgatadi ☑: turli formatlarda (JSON, tables) javob berish, roles, <EOS> token, kursatmaga amal qilish.

SFT nimani urgatmaydi ✖: yangi bilimni urgatmaydi, qaysi javob kuproq afzalligini bilmaydi. Havfli javobdan chiqib ketishni bilmaydi.

SFT orqali nimalar qilamiz:

Ability	Definition
Instruction following	Userni kursatmalariga amal qilish
Response formatting	Javoblarni kerakli formatda yozish
Question answering	Savollarga mos javob berish
Summarization	Uzun matnni kesib, gapni jamlab berish
Translation	Bir tildan boshqa tilga tarjima qilish
Code generation	Code yozish
Conversation	Suhbat davomida theme ga mos javob berish
Response style	Qisqa yoki rasmiy javob berish

Pre-training vs. SFT:

Feature	Pre-training	SFT
Asosiy maqsad	Til va matn qoidalarini urganish	Instruction following ability ni rivojlantirish
Dataset	Katta hajmdagi matnlar	Instruction-response juftliklari
Input	X	Instruction
Target	y	Expected response
Loss	MSE, Cross-entropy loss	Next-token Cross-entropy loss
Model	Base-model	Instruction-tuned model
Natija	Keyingi tokenni bashorat qilibla ketishga urgatilingan model	Kursatmalarga javob berishga moslashtirilgan model

SFT pipeline:

Training data (instructions) → SFT Tokenization → Model Predicts next token → Loss calculation (accuracy of predicted token and actual token) → Backpropagation (Gradients) → Update parameters → The sequence repeats many times

Behavioral Cloning

Behavioral cloning: AI ni insonning harakatlarini misollar orqali urgatish usuli. Model kuzatib va unga mos harakatni urganib, shunga uxshash vaziyatlarda
shu harakatni bajarishga harakat qiladi

Boshqa sohalarda: Robototexnikada robotni boshqarish, avtonom vehiclelarda haydashni urganish

LLM da: Modelga savol va unga mos javoblari beriladi. Model shu examplelar orqali kerakli javob berish va kursatmalarga amal qilishni urganadi.

Natijada, savolga tugridan-tugri va keraksiz suzlarni qushmagan holda qisqa tushunarli javob beradigon model chiqadi

Teacher Forcing

Teacher Forcing (TF): modelni training jarayonida unga oldingi haqiqiy tokenlarni berib, keyingi tokenni bashorat qilishni urgatish usuli. model savol va togri javob misollari orqali uqitiladi. Modelni hato bashorat qilgan tokenini tugri token ga replace qiladi

Nima uchun kerak: Model kuplab misollar orqali qonuniyatlarni urganadi va uqitish tugab, tugri tokenlar berilmasa ham, javobni uzi chiqarishga harakat qiladi.

Muhim: Teacher Forcing malumotlar leakage bulishi emas. u uqitish uchun ishlatiladi, test paytida esa model weightlarni update qilmasi kerak.

Plus/Minus tomonlari: training jarayon tezroq buladi va GPU da parallel ishlash imkoni bor. Lekin, inference da hatoliklar paydo bulish ehtimoli kup. Shu muammo sababli erkin generation yaxshi qila ololmaydi.

SFT - Instruction

Asosiy 3 ta style bor:

1. Question – Answer: faqat savol va javob kurinishida buladi
2. Instruction question – answer: modelga nima qilish kerakligi aytiladi, savol beriladi va javob olinadi. Lekin input har doim ham bulishi shart emas.
3. Message: role, Q&A, hammasi buladi va xabarlar orqali tuziladi (system)

Dataset Formats

Format	Tuzilishi	Multi-turn	Qayerda uchraydi
Alpaca	Instruction, input, output	No	Stanford alpaca, alpaca-cleaned
ShareGPT	Conversations: [{from, value}]	Yes	Vicuna, kup community datalar
Messages	Messages: [{role, content}]	Yes	HF chat template, TRL – STANDARD
Prompt-completion	Prompt, completion	Qisman	TRL: loss faqat completionda